

Periodic Orbits and Dynamical Structures in the Circular Concentric Restricted Four-Body Problem

Gunnar Montseny Gens*

University of Colorado Boulder, Boulder, Colorado 80303

I. Introduction

THE Circular Concentric Restricted Four-Body Problem (CCR4BP) builds upon the Circular Restricted Three-Body Problem (CR3BP) by introducing the gravitational influence of a third massive primary moving on a circular orbit. This additional body introduces a time-periodic perturbation to the system, resulting in a non-autonomous Hamiltonian Dynamical System (HDS). The CCR4BP serves as a useful intermediate model between the CR3BP and the general Restricted N-Body Problem (RNBP). In particular, it provides a valuable framework for studying the motion of spacecraft in multi-body environments such as the Jupiter–Europa–Ganymede and Saturn–Janus–Epimetheus, where the presence of an additional primary significantly alters the dynamical structure.

This work focuses on the analysis of periodic motion within the planar CCR4BP, with an emphasis on the existence and stability of periodic orbits near Lagrange points. Using both Lagrangian and Hamiltonian formulations, the equations of motion and associated state transition matrix are derived and numerically integrated to investigate the system’s dynamical behavior. Periodic orbits are computed using simple shooting methods and perpendicular crossings, and their stability is assessed through the eigenvalues of the monodromy matrix.

II. Model Description

The Circular Concentric Restricted Four-Body Problem (CCR4BP) describes the motion of a massless particle under the gravitational influence of three massive primaries. In this model, the primaries are assumed to move on prescribed circular, concentric orbits about a common center of mass. As a result, the system is non-autonomous and governed by a time-periodic Hamiltonian.

In this work, we restrict our attention to the planar CCR4BP, assuming that all bodies lie in the same plane. The three primaries have masses m_1 , m_2 , and m_3 , and the massless body (referred to as the *spacecraft* or *particle*) does not influence their motion. We assume that $m_1 \gg m_2, m_3$, so that m_2 and m_3 revolve around m_1 in coplanar circular orbits of radii r_{12} and r_{13} , respectively. The motion of m_2 and m_3 is prescribed and mutually uncoupled, meaning that they do not gravitationally influence each other. Although this assumption does not satisfy the full N -body equations of motion, it provides a useful approximation for certain physical systems, such as the Jupiter–Europa–Ganymede system [1].

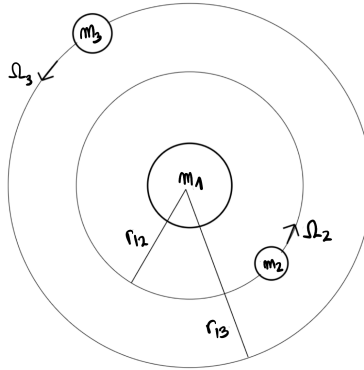


Fig. 1 Schematic representation of the CCR4BP in an inertial frame. The primaries m_2 and m_3 revolve around the dominant mass m_1 with angular velocities Ω_2 and Ω_3 , respectively. Image from [1].

*Ann and H.J. Smead Department of Aerospace Engineering Sciences, University of Colorado Boulder, 3775 Discovery Drive, Boulder, CO 80303, USA

From Kepler's third law, the angular velocities of m_2 and m_3 around m_1 are given by

$$\Omega_2 = \sqrt{\frac{G(m_1 + m_2)}{r_{12}^3}}, \quad \Omega_3 = \sqrt{\frac{G(m_1 + m_3)}{r_{13}^3}}. \quad (1)$$

These are related to the orbital periods by $\Omega_i = 2\pi/T_i$, for $i = 2, 3$. To simplify the equations of motion, we introduce the mass parameters:

$$\mu = \frac{m_2}{m_1 + m_2}, \quad \mu_3 = \frac{m_3}{m_1 + m_2}, \quad (2)$$

and adopt the standard non-dimensionalization

$$G(m_1 + m_2) = 1, \quad r_{12} = 1. \quad (3)$$

Under this normalization, Eq. (1) reduces to:

$$\Omega_2 = 1, \quad \Omega_3 = \frac{T_2}{T_3}. \quad (4)$$

To analyze the dynamics, we introduce a synodic (rotating) reference frame $\mathcal{R}$ with angular velocity Ω_2 . In this frame, the primaries m_1 and m_2 remain fixed along the x -axis, and the origin is located at their barycenter. Their positions are given by

$$\vec{r}_1 = (-\mu, 0), \quad \vec{r}_2 = (1 - \mu, 0), \quad (5)$$

while the position of the third primary becomes time-dependent:

$$\vec{r}_3(t) = (-\mu + r_{13} \cos \theta_3(t), r_{13} \sin \theta_3(t)), \quad (6)$$

where $\theta_3(t) = (\Omega_3 - 1)t + \theta_0$, and θ_0 denotes the initial phase.

Let $\vec{r} = (x, y)$ denote the position of the massless particle in this frame. We define the relative position vectors

$$\vec{\eta}_i = \vec{r} - \vec{r}_i, \quad i \in \{1, 2, 3\}, \quad (7)$$

with the corresponding distances:

$$\begin{aligned} \eta_1(x, y) &= \sqrt{(x + \mu)^2 + y^2}, \\ \eta_2(x, y) &= \sqrt{(x - 1 + \mu)^2 + y^2}, \\ \eta_3(x, y, t) &= \sqrt{(x + \mu - r_{13} \cos \theta_3(t))^2 + (y - r_{13} \sin \theta_3(t))^2}. \end{aligned}$$

See Figure 2 for the geometric configuration of the CCR4BP in the synodic frame $\mathcal{R}$. The presence of the third primary introduces a time-dependent perturbation through $\theta_3(t)$, making the system explicitly time-periodic. The forcing period is given by

$$T_p = \frac{2\pi}{|\Omega_3 - \Omega_2|} = \frac{2\pi}{|\Omega_p - 1|}, \quad (8)$$

which plays a central role in the existence of periodic and quasi-periodic solutions in the CCR4BP.

III. Theoretical Framework

This section presents the governing equations of motion of the CCR4BP in both Lagrangian and Hamiltonian formulations. The dynamics are expressed in the synodic reference frame $\mathcal{R}$, in which the primaries m_1 and m_2 remain fixed. We derive the equations in a form suitable for numerical integration and introduce the associated equations of motion for the State Transition Matrix (STM).

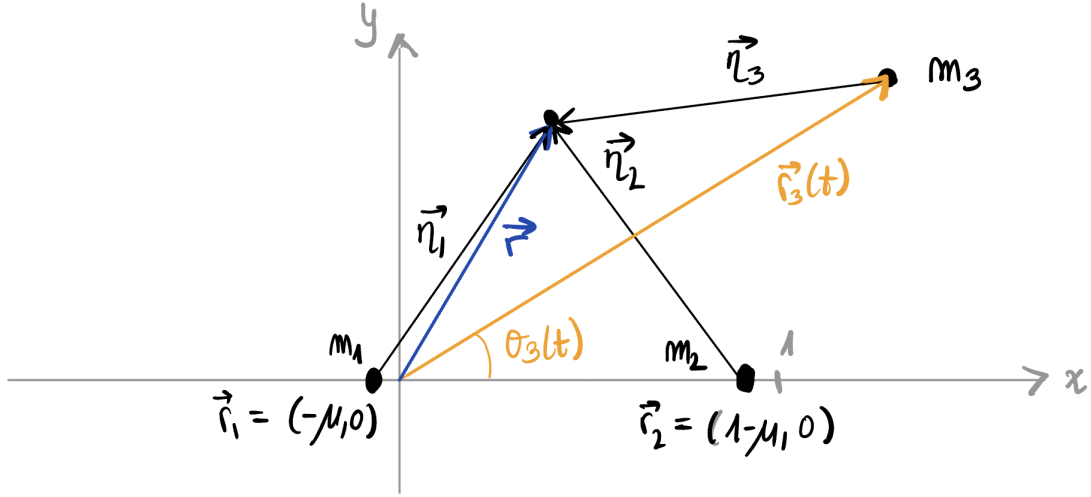


Fig. 2 Geometric configuration in the synodic frame $\mathcal{R}$. The vectors $\vec{r}_i$ represent the relative position of the spacecraft with respect to each primary.

A. Lagrangian Equations of Motion

Consider the RNBP, where $N - 1$ bodies influence the motion of a massless body. The equations of motion can be derived with the Lagrangian $\mathcal{L} = T + U$, where T is the kinetic energy of the massless object, and U is the acceleration potential. The acceleration potential must satisfy:

$$\ddot{\vec{r}} = \frac{\partial}{\partial \vec{r}} U.$$

First, we calculate the kinetic energy of the particle. We use the Transport Theorem:

$$T = \frac{1}{2} \left((\dot{x} - \dot{y})^2 + (\dot{y} + \dot{x})^2 \right) \quad (9)$$

Next, we find the acceleration potential. The acceleration potential of the CR3BP is:

$$U_{C3BP}(\vec{r}) = G \left(\frac{m_1}{\eta_1(\vec{r})} + \frac{m_2}{\eta_2(\vec{r})} \right). \quad (10)$$

Through the normalization presented in Eq 3, the following equations hold:

$$Gm_1 = 1 - \mu, \quad Gm_2 = \mu, \quad Gm_3 = 1 - \mu_3.$$

In our case, we have that $U_{CCR4BP} = U_{C3BP} + U_3$, where U_3 is a disturbing potential associated with the third primary that has the following expression:

$$U_3(\vec{r}, t) = \frac{\mu_3}{\eta_3(\vec{r}, t)} - \mu_3 \frac{\vec{r}_1(\vec{r}) \cdot \vec{r}_3(\vec{r}, t)}{r_{13}^3}, \quad (11)$$

where the first summand is known as the direct term, and the second summand is known as the indirect term. The indirect term is added to account for the fact that the origin is not located at the barycenter of the three massive bodies. It is important to note that this disturbing potential has an explicit time dependence. The full expression for the Lagrangian can now be written:

$$\begin{aligned} \mathcal{L}(x, y, \dot{x}, \dot{y}, t) = & \frac{1}{2} \left((\dot{x} - \dot{y})^2 + (\dot{y} + \dot{x})^2 \right) \\ & + \frac{1 - \mu}{\eta_1(x, y)} + \frac{\mu}{\eta_2(x, y)} + \frac{\mu_3}{\eta_3(x, y, t)} - \mu_3 \frac{x \cos \theta_3(t) + y \sin \theta_3(t)}{r_{13}^2}. \end{aligned} \quad (12)$$

The Lagrangian equations of motion are:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x}, \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = \frac{\partial \mathcal{L}}{\partial y}, \quad (13)$$

where we have considered $\vec{q} = (x, y)^T$ as our generalized coordinates, and $\dot{\vec{q}} = (\dot{x}, \dot{y})^T$ as their time derivatives. After many algebraic manipulations, we obtain the equations of motion of the CCR4BP in Lagrangian coordinates:

$$\ddot{x} = 2\dot{y} + x - \frac{1-\mu}{\eta_1^3}(x+\mu) - \frac{\mu}{\eta_2^3}(x-1+\mu) - \frac{\mu_3}{\eta_3^3}(x+\mu-r_{13}\cos\theta_3(t)) - \mu_3 \frac{\cos\theta_3(t)}{r_{13}^2} \quad (14)$$

$$\ddot{y} = -2\dot{x} + y - \frac{1-\mu}{\eta_1^3}y - \frac{\mu}{\eta_2^3}y - \frac{\mu_3}{\eta_3^3}(y-r_{13}\sin\theta_3(t)) - \mu_3 \frac{\sin\theta_3(t)}{r_{13}^2} \quad (15)$$

B. Hamiltonian Equations of Motion

First, we define the momentas:

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = \dot{x} - y, \quad p_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = \dot{y} + x, \quad (16)$$

which allows us to relate the Hamiltonian coordinates to the Lagrangian coordinates. Let $\vec{p} = (p_x, p_y)^T$ be the vector containing the momentas. Then we have that the Hamiltonian of this system can be calculated with the following expression:

$$\mathcal{H}(\vec{q}, \vec{p}, t) = \vec{p} \cdot \dot{\vec{q}} - \mathcal{L}(\vec{q}, \dot{\vec{q}}(\vec{q}, \vec{p}), t). \quad (17)$$

As a result, we have that:

$$\mathcal{H}(x, y, p_x, p_y, t) = \frac{1}{2} (p_x^2 + p_y^2) + p_x y - p_y x - \frac{1-\mu}{\eta_1(x, y)} - \frac{\mu}{\eta_2(x, y)} - \frac{\mu_3}{\eta_3(x, y, t)} + \mu_3 \frac{x \cos \theta_3(t) + y \sin \theta_3(t)}{r_{13}^2}. \quad (18)$$

The Hamiltonian equations of motion are:

$$\dot{x} = \frac{\partial \mathcal{H}}{\partial p_x}, \quad \dot{y} = \frac{\partial \mathcal{H}}{\partial p_y}, \quad \dot{p}_x = -\frac{\partial \mathcal{H}}{\partial x}, \quad \dot{p}_y = -\frac{\partial \mathcal{H}}{\partial y}. \quad (19)$$

For our system, they take the following form:

$$\begin{aligned} \dot{x} &= p_x + y \\ \dot{y} &= p_y - x \\ \dot{p}_x &= p_y - \frac{1-\mu}{\eta_1^3}(x+\mu) - \frac{\mu}{\eta_2^3}(x-1+\mu) - \frac{\mu_3}{\eta_3^3}(x+\mu-r_{13}\cos\theta_3(t)) - \mu_3 \frac{\cos\theta_3(t)}{r_{13}^2} \\ \dot{p}_y &= -p_x - \frac{1-\mu}{\eta_1^3}y - \frac{\mu}{\eta_2^3}y - \frac{\mu_3}{\eta_3^3}(y-r_{13}\sin\theta_3(t)) - \mu_3 \frac{\sin\theta_3(t)}{r_{13}^2} \end{aligned} \quad (20)$$

From now on, we take the state of the system to be $\vec{\chi} = (x, y, p_x, p_y)^T$, meaning that $(\vec{\chi}, \mathcal{H}(\vec{\chi}, t))$ forms a time-varying HDS. Specifically, it is time periodic with period T_p . Let the Hamiltonian equations of motion in Eq 20 be abbreviated as:

$$\dot{\vec{\chi}} = \vec{f}(\vec{\chi}, t). \quad (21)$$

C. Equations of Motion for the State Transition Matrix

We define the State Transition Matrix (STM) as

$$\Phi(t, t_0; \vec{\chi}_0) = \frac{\partial \vec{\chi}}{\partial \vec{\chi}_0} \in \mathbb{R}^{4 \times 4}, \quad (22)$$

which describes how infinitesimal perturbations in the initial state evolve along a trajectory. We also define the symplectic identity matrix J :

$$J = \begin{bmatrix} 0_{2 \times 2} & I_{2 \times 2} \\ I_{2 \times 2} & 0_{2 \times 2} \end{bmatrix} \in \mathbb{R}^{4 \times 4} \quad (23)$$

The dynamics of the STM are governed by:

$$\dot{\Phi} = A\Phi, \quad \Phi(0) = I_{4 \times 4}, \quad (24)$$

where $A \in \mathbb{R}^{4 \times 4}$ is known as the dynamics matrix. The dynamics matrix is the Jacobian of the Hamiltonian vector field:

$$A = \frac{\partial \vec{f}}{\partial \vec{\chi}} = J \frac{\partial^2 \mathcal{H}}{\partial \vec{\chi}^2} \quad (25)$$

Let $x^*, y^*, t^* \in \mathbb{R}$ be fixed. In the CCR4BP, the dynamics matrix has the following form:

$$A(x^*, y^*, t^*) = \begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -\mathcal{H}_{xx} & -\mathcal{H}_{xy} & 0 & 1 \\ -\mathcal{H}_{yx} & -\mathcal{H}_{yy} & -1 & 0 \end{bmatrix}. \quad (26)$$

To compute the lower left submatrix of A , we consider $\rho_{a,b}(x, y) = \sqrt{(x-a)^2 + (y-b)^2}$, where $a, b \in \mathbb{R}$. Then:

$$\begin{cases} \frac{\partial}{\partial x^2} \left(\frac{1}{\rho_{a,b}} \right) (x^*, y^*) = -\frac{1}{\rho_{a,b}^3(x^*, y^*)} + 3\frac{(x^* - a)^2}{\rho_{a,b}^5(x^*, y^*)} \\ \frac{\partial}{\partial y^2} \left(\frac{1}{\rho_{a,b}} \right) (x^*, y^*) = -\frac{1}{\rho_{a,b}^3(x^*, y^*)} + 3\frac{(y^* - b)^2}{\rho_{a,b}^5(x^*, y^*)} \\ \frac{\partial}{\partial x \partial y} \left(\frac{1}{\rho_{a,b}} \right) (x^*, y^*) = \frac{\partial}{\partial y \partial x} \left(\frac{1}{\rho_{a,b}} \right) (x^*, y^*) = 3\frac{(x^* - a)(y^* - b)}{\rho_{a,b}^5(x^*, y^*)} \end{cases}$$

Note that $p_{-\mu,0} = \eta_1$, $\rho_{1-\mu,0} = \eta_2$ and $\rho_{x_3,y_3} = \eta_3$. As a result, the remaining elements in the dynamics matrix are:

$$\mathcal{H}_{kl}(x^*, y^*, t^*) = -(1-\mu) \frac{\partial}{\partial k \partial l} \left(\eta_1^{-1} \right) (x^*, y^*) - \mu \frac{\partial}{\partial k \partial l} \left(\eta_2^{-1} \right) (x^*, y^*) - \mu_3 \frac{\partial}{\partial k \partial l} \left(\eta_3^{-1} \right) (x^*, y^*, t^*), \quad (27)$$

where $k, l \in \{x, y\}$.

D. Numerical Integration

Consider the Hamiltonian state $\vec{\chi} = (x, y, p_x, p_y)^T$, and STM Φ . Then the combined state we numerically integrate is:

$$\vec{X} = \begin{bmatrix} \vec{\chi} \\ \text{vec}\Phi \end{bmatrix} \in \mathbb{R}^{20},$$

where $\text{vec}\Phi$ is the vectorized version of the STM. The dynamics of both the state and the STM are governed by the following equation:

$$\dot{\vec{X}} = \vec{F}(\vec{X}), \quad (28)$$

where $\vec{F} \in \mathbb{R}^{20}$ is a vector field that can be determined explicitly with Eqs 20, 24. In this work, we numerically integrate $\vec{F}$ with ode45, with $RelTol = 1e-13$ and $AbsTol = 1e-13$.

E. Stability and Topology of the CCR4BP

Before introducing specific orbits in the CCR4BP, it is important to characterize the structure of its phase space. The phase space of the CCR4BP is homeomorphic to $\mathbb{R}^4 \times \mathbb{S}^1$. This follows from the fact that $\vec{\chi} \in \mathbb{R}^4$, and that the variable $\tau = t \bmod T_p$ can be introduced to describe the phase of the third primary. As a result, $\tau \in \mathbb{R}/T_p\mathbb{Z} \cong \mathbb{S}^1$, and the full state of the system can be uniquely represented by

$$(x, y, p_x, p_y, \tau) \in \mathbb{R}^4 \times \mathbb{S}^1.$$

From now on, we define $\vec{\xi} = (\vec{\chi}, \tau)$, allowing the CCR4BP to be written as an autonomous HDS in an extended phase space. In this framework, different types of trajectories correspond to different subsets of the phase space. In particular, we focus on periodic orbits and their stability properties. Since the CCR4BP is a time-periodic HDS with forcing period T_p , the simplest solutions are periodic orbits whose periods are commensurate with T_p . We define a periodic orbit with period T as:

$$\mathcal{P}(T) = \{\vec{\xi}(t) \in \mathbb{R}^4 \times \mathbb{S}^1 \mid \vec{\xi}(t+T) = \vec{\xi}(t) \forall t \in \mathbb{R}\}, \quad (29)$$

where $T \in T_p\mathbb{Z}$. Topologically, we have that $\mathcal{P} \cong \mathbb{S}^1$. We will frequently use $\mathcal{P}(T, L_i)$ to refer to a periodic orbit of period T about Lagrange Point i , or $\mathcal{P}(T, \text{Planet})$ to refer to a periodic orbit of period T about a planet. The stability of a periodic orbit can be analyzed through its monodromy matrix, which is defined as:

$$\Phi_M(\mathcal{P}(T)) = \Phi(T, 0) \in \mathbb{R}^{4 \times 4}, \quad (30)$$

where $\Phi(T, 0)$ is the STM of periodic orbit $\mathcal{P}$ evaluated at period T . Let $\lambda \in \mathbb{C}$ be an eigenvalue of the monodromy matrix. Then:

- $|\lambda| > 1 \implies$ unstable mode (exponential growth)
- $|\lambda| < 1 \implies$ stable mode (exponential decay)
- $|\lambda| = 1, \lambda \neq 1 \implies$ neutrally stable oscillatory mode (center manifold)
- $\lambda = 1 \implies$ singular mode

Due to the symplectic structure of the Hamiltonian system, if $\lambda \in \mathbb{C}$ is an eigenvalue of the monodromy matrix, then λ^{-1} , $\bar{\lambda}$, and $\bar{\lambda}^{-1}$ are also eigenvalues. Next, let $\lambda = \rho e^{i\theta}$, where $\rho \in \mathbb{R}$ and $\theta \in [0, 2\pi)$. When $\rho = 1$, the eigenvalues lie on the unit circle and correspond to neutrally stable oscillatory modes. If θ is incommensurate with π , the motion in the vicinity of the periodic orbit is quasi-periodic. In this case, the dynamics can be described by invariant tori in the extended phase space. Since a periodic orbit already corresponds to a one-dimensional invariant set ($\mathbb{S}^1$), the presence of additional independent frequencies increases the dimensionality of the invariant set:

- One additional incommensurate frequency leads to a two-dimensional quasi-periodic orbit, homeomorphic to $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$.
- Two independent incommensurate frequencies lead to a three-dimensional quasi-periodic orbit, homeomorphic to $\mathbb{T}^3 = \mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{S}^1$.

IV. An application to the Jupiter - Europa - Ganymede system

A. System Description

The Jupiter–Europa–Ganymede (JEG) system is well approximated by the CCR4BP due to its strong mass hierarchy ($m_1 \gg m_2, m_3$), which allows Europa and Ganymede to be modeled as independent circular orbits about Jupiter. An important feature of this system is the *Laplace resonance*, which enforces an approximate 2:1 commensurability between the orbital periods of Europa and Ganymede. In the synodic frame rotating with Europa, this results in a time-periodic perturbation induced by Ganymede with forcing period $T_p \approx 4\pi$.

Table 1 Parameters for the Jupiter–Europa–Ganymede CCR4BP model.

Parameter	Symbol	Value (Dimensional)	Value (Non-dimensional)
<i>Mass Parameters</i>			
Jupiter mass	m_1	1.898×10^{27} kg	$1 - \mu$
Europa mass	m_2	4.780×10^{22} kg	μ
Ganymede mass	m_3	1.482×10^{23} kg	μ_3
Mass ratio	μ	–	2.529×10^{-5}
Perturbing mass ratio	μ_3	–	7.807×10^{-5}
<i>Europa's Motion</i>			
Jupiter–Europa distance	r_{12}	670900 km	1
Europa orbital period	T_2	3.551 days	$2\pi \approx 6.283$
Europa angular rate	Ω_2	$4.23^\circ/\text{hr}$	1
<i>Ganymede's Motion</i>			
Jupiter–Ganymede distance	r_{13}	1070400 km	1.595
Ganymede orbital period	T_3	7.155 days	12.569
Ganymede angular rate	Ω_3	$2.10^\circ/\text{hr}$	0.496
Initial phase	θ_0	0°	0
<i>Time-Periodic Perturbation</i>			
Forcing period	T_p		$12.475 \approx 4\pi$

It is important to note that in the upcoming plots, Lagrange Points are shown. They correspond to the Lagrange Points of the Jupiter - Europa system modeled with the CR3BP.

B. Numerical Verification

To verify the numerical integration, we evaluate whether the propagated State Transition Matrix (STM) preserves the symplectic structure of the Hamiltonian system by monitoring the Frobenius norm of the symplectic error, $\|\Phi(t)^T J \Phi(t) - J\|_F$. Figure 3 shows the time evolution of this quantity for a randomly chosen initial condition and two different tolerance settings. In both cases, the symplectic error remains small and bounded, and decreases as the integration tolerances are tightened, indicating that the observed error is primarily due to computational limitations rather than inconsistencies in the equations of motion.

To further verify the behavior in a more relevant setting, the same symplectic check is performed along a computed periodic orbit about L_4 of the JEG system, as shown in Fig.4. In this case, the symplectic error remains below 10^{-11} throughout the integration, demonstrating that the numerical scheme preserves the Hamiltonian structure to a very high degree of accuracy along trajectories of interest. This provides confidence in the reliability of the integrator for the periodic orbit computations presented in this work.

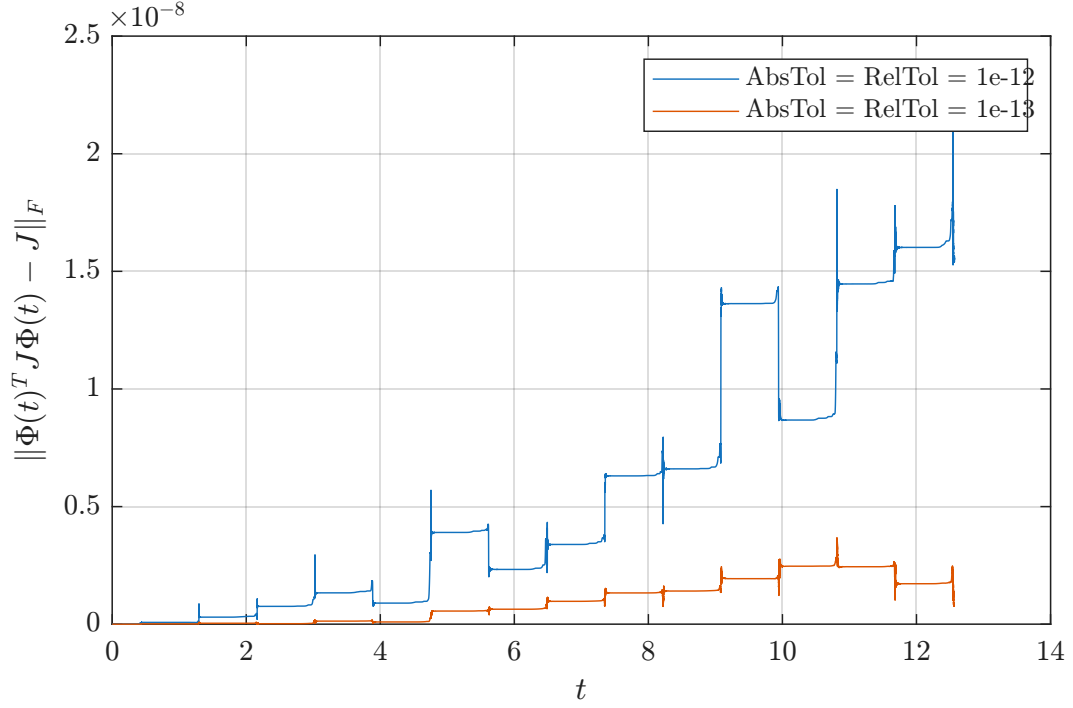


Fig. 3 Frobenius norm of the symplectic error as a function of time for a randomly selected initial condition. Results are shown for two tolerance settings.

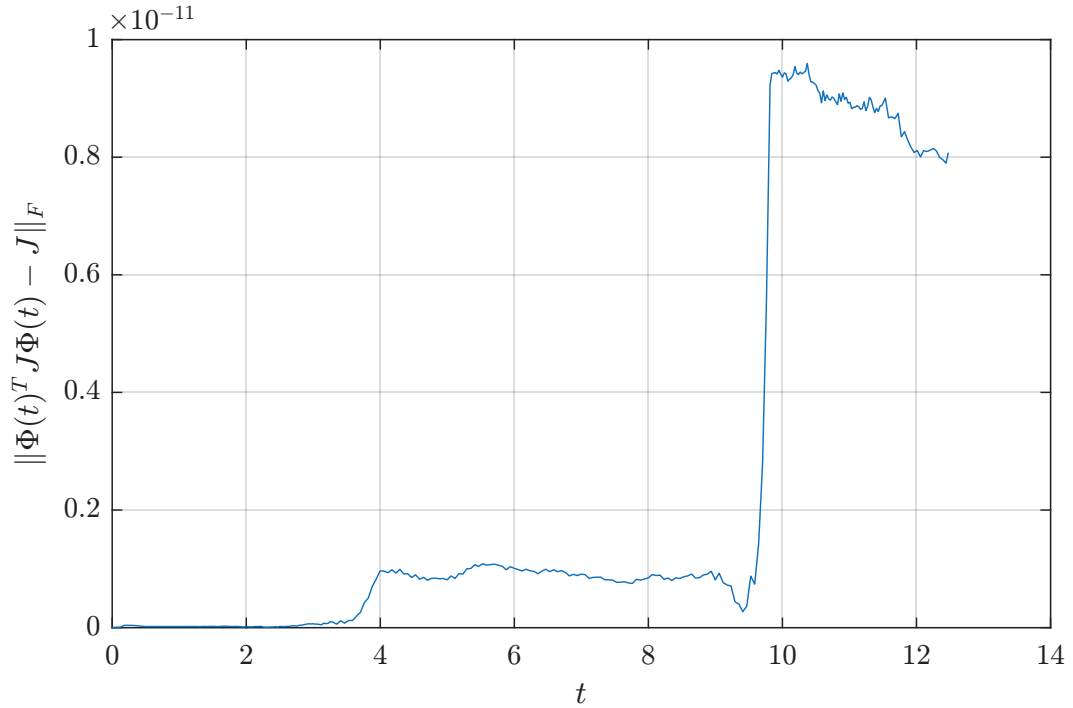


Fig. 4 Frobenius norm of the symplectic error as a function of time for a periodic orbit about L_4 in the JEG system.

C. Lyapunov-like Periodic Orbits around L_1

To construct Lyapunov-like periodic orbits about L_1 in the CCR4BP, we employ a perpendicular crossing approach. This method builds on the structure of Lyapunov orbits in the CR3BP, where the linearized motion about L_1 has a characteristic period $T_{L_1, \text{lin}} \approx \pi$ [1]. In the CCR4BP, this characteristic timescale is only weakly affected by the presence of the third body, allowing us to use it as a reference [1]. For the JEG system, the angular rates satisfy $\Omega_2 \approx 2\Omega_3$, which leads to a forcing period $T_p \approx 4\pi$. As a result, the linear oscillation period about L_1 satisfies

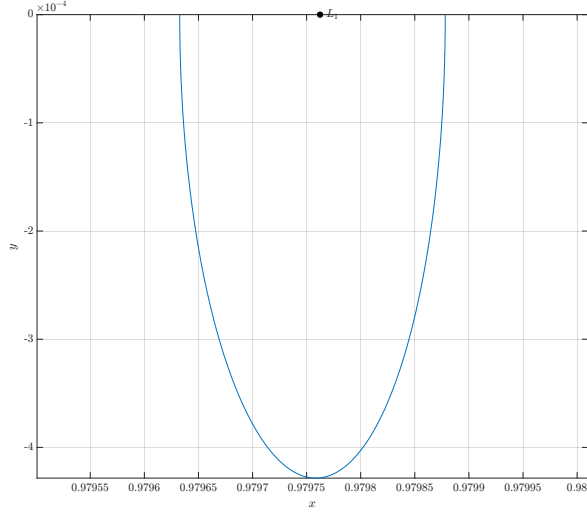
$$T_{L_1, \text{lin}} \approx \frac{1}{4} T_p.$$

This relation plays a key role in constructing periodic orbits, as it determines how many oscillations are required before the trajectory closes under the time-periodic forcing. In a time-periodic system, a periodic orbit must be commensurate with the forcing period. That is, after p oscillations and q forcing cycles, the trajectory must return to its initial state, yielding the condition

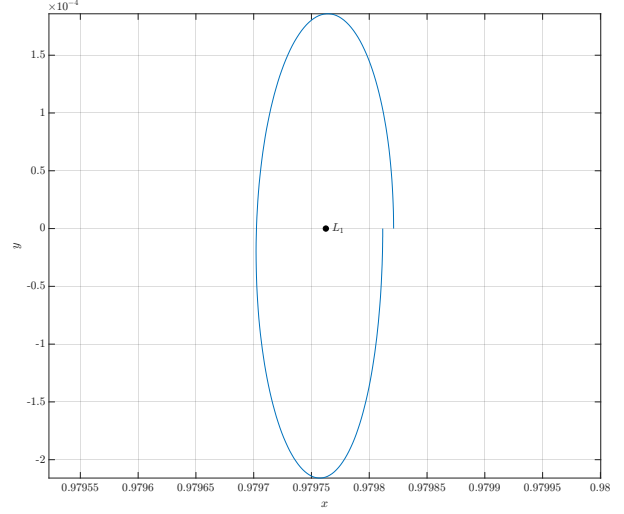
$$T_{\text{PO}} = p T_{L_1, \text{lin}} = q T_p.$$

In the JEG system, the smallest such orbit corresponds to $q = 1$, leading to $p = 4$. Therefore, the trajectory must complete four oscillations about L_1 before closing. The perpendicular crossing condition is defined by $y = 0, \dot{x} = 0$, which ensures that the trajectory intersects the x -axis orthogonally. Starting from an initial condition near L_1 , the trajectory is propagated forward in time until it satisfies these conditions. This process is repeated, enforcing the same conditions at each return to the x -axis.

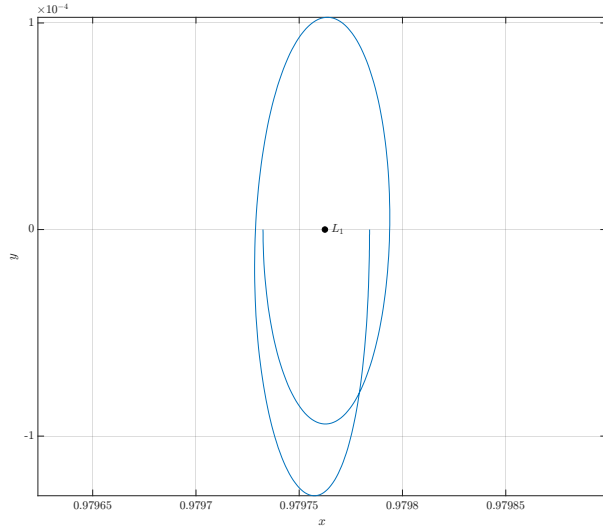
After eight such crossings, corresponding to one full forcing period T_p , the trajectory provides a good initial guess for a periodic orbit. This initial guess is then refined using a Newton-based shooting method to obtain a Lyapunov-like periodic orbit about L_1 with period T_p . In practice, however, we have not been able to obtain all eight perpendicular crossings for trajectories about L_1 . While the first few crossings can be computed reliably, the motion becomes highly sensitive in this region, and the trajectory quickly departs from the vicinity of L_1 . Figure 5 contains the first three perpendicular crossings, and shows how a fourth perpendicular crossing has not been found.



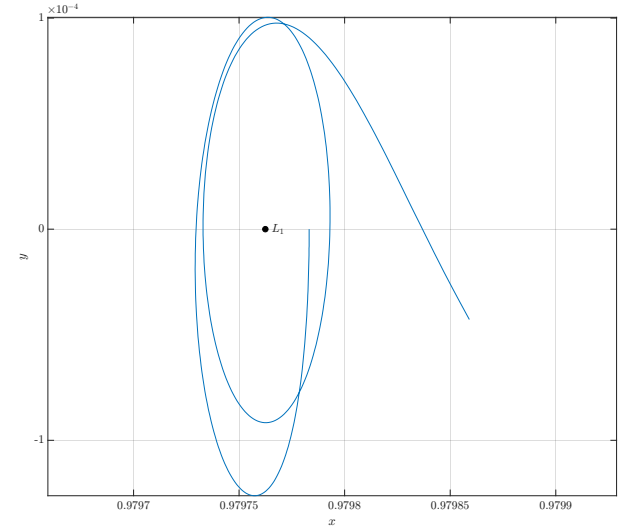
(a) First perpendicular crossing.



(b) Second perpendicular crossing.



(c) Third perpendicular crossing.



(d) Fourth perpendicular crossing.

Fig. 5 First four perpendicular crossings needed to generate a Lyapunov-like Periodic Orbit with period T_p about $L_1, \mathcal{P}(T_p, L_1)$.

D. Periodic Orbit about L_4

Since the dynamics in the vicinity of L_4 are predominantly stable, we have computed a periodic orbit about L_4 , denoted $\mathcal{P}(T_p, L_4)$, using a standard single-shooting algorithm. The eigenvalues of the corresponding monodromy matrix, $\Phi_M(\mathcal{P}(T_p, L_4))$, lie on the unit circle and exhibit two distinct frequencies. As a result, the motion in the neighborhood of $\mathcal{P}(T_p, L_4)$ is stable and quasi-periodic, specifically confined to a 3D QPO homeomorphic to $\mathbb{T}^3$.

Table 2 Initial conditions and period of $\mathcal{P}(T_p, L_4)$.

x_0	y_0	$\dot{x}_0$	$\dot{y}_0$	T
0.490724316497966	0.875755357407918	0.009978755940005	0.001810383081939	$T_p \approx 12.475$

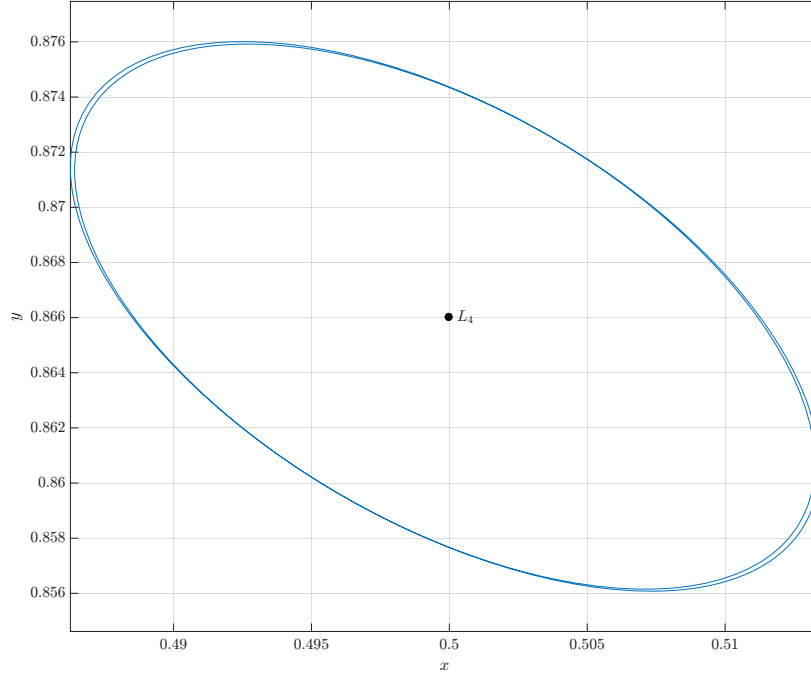


Fig. 6 Periodic Orbit about $L_4, \mathcal{P}(T_p, L_4)$.

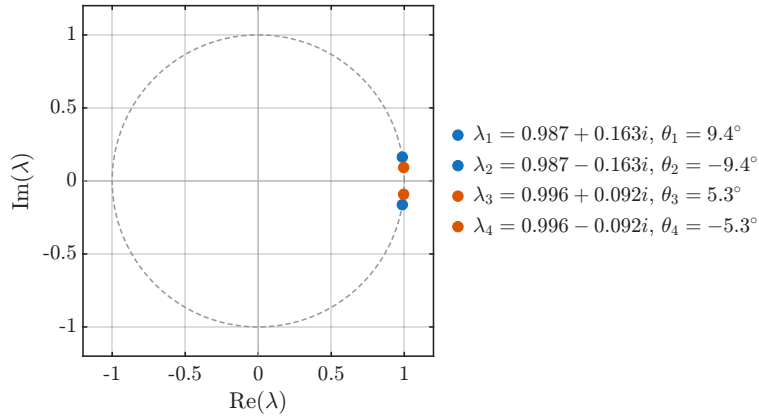


Fig. 7 Eigenvalues of the monodromy matrix $\Phi_M(\mathcal{P}(T_p, L_4))$. Their magnitude is approximately 1.

E. Periodic Orbit about L_5

We have also computed a periodic orbit about L_5 , denoted $\mathcal{P}(T_p, L_5)$, using a standard single-shooting algorithm. The motion in the vicinity of $\mathcal{P}(T_p, L_5)$ is stable and topologically analogous to the motion near $\mathcal{P}(T_p, L_4)$.

Table 3 Initial conditions and period of $\mathcal{P}(T_p, L_5)$.

x_0	y_0	$\dot{x}_0$	$\dot{y}_0$	T
0.490724316502206	-0.875755357405315	-0.009978755939699	0.001810383082140	$T_p \approx 12.475$

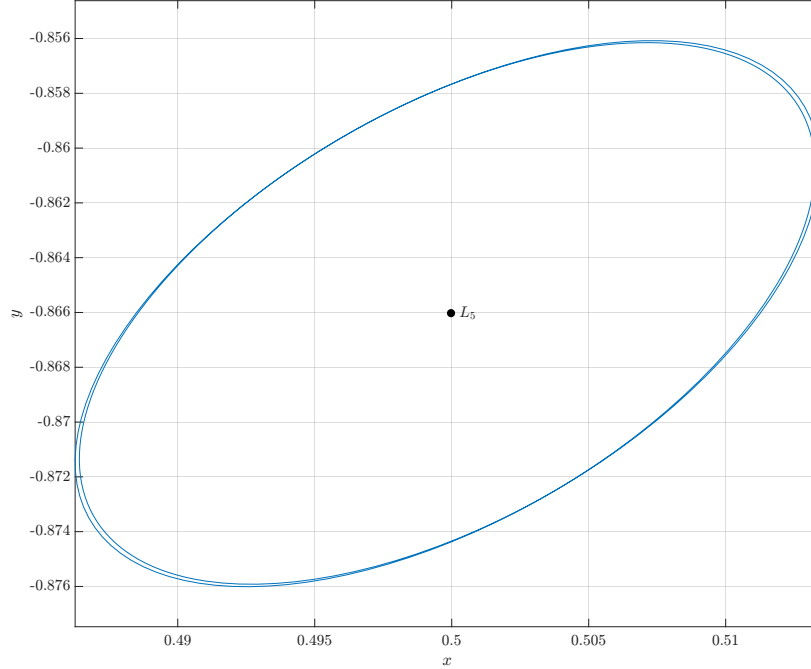


Fig. 8 Periodic Orbit about $L_5, \mathcal{P}(T_p, L_5)$.

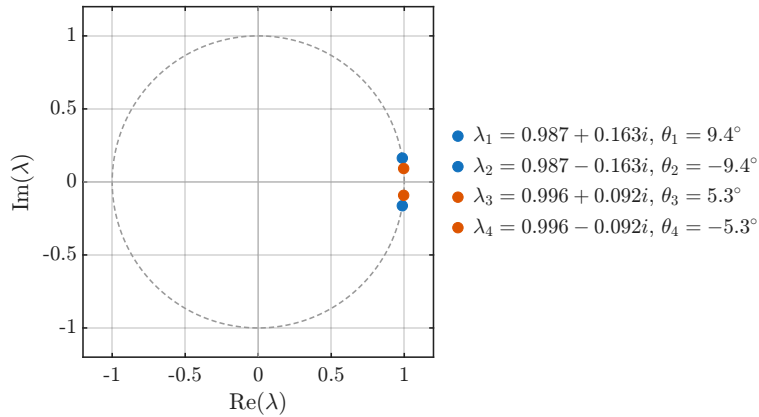


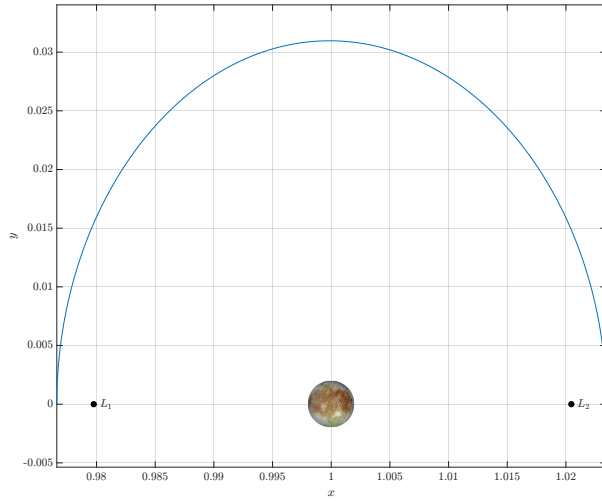
Fig. 9 Eigenvalues of the monodromy matrix $\Phi_M(\mathcal{P}(T_p, L_5))$. Their magnitude is approximately 1.

F. Periodic Orbit around Europa

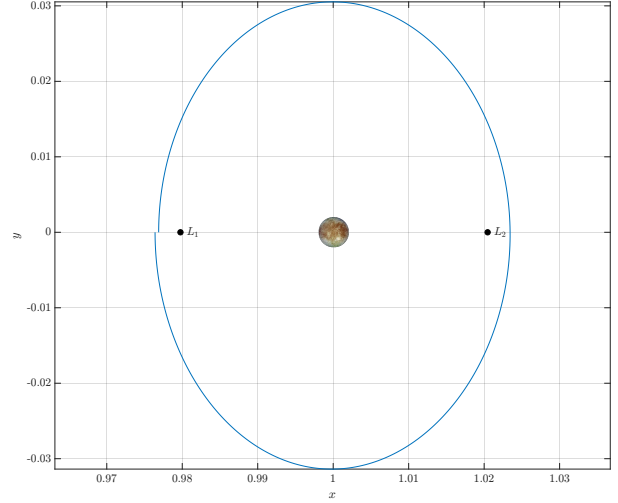
We have computed a periodic orbit about Europa, denoted $\mathcal{P}(T_p, \text{Europa})$, using the perpendicular crossings method presented earlier. The eigenvalues of the corresponding monodromy matrix lie on the unit circle and exhibit two distinct frequencies. Consequently, the motion in the vicinity of $\mathcal{P}(T_p, \text{Europa})$ is stable and quasi-periodic, specifically confined to a 3D QPO homeomorphic to $\mathbb{T}^3$.

Table 4 Initial conditions and period of $\mathcal{P}(T_p, \text{Europa})$,

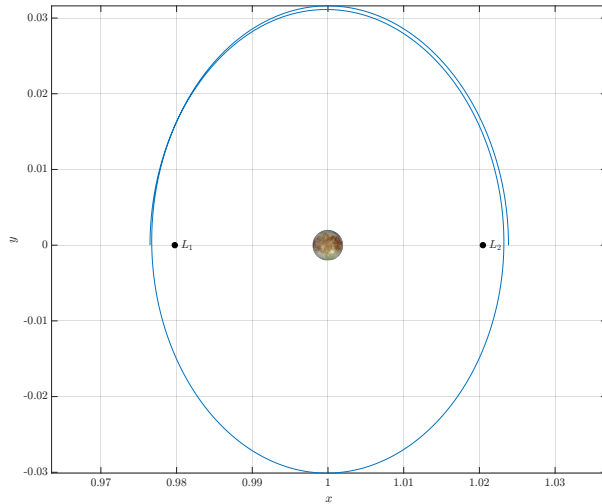
x_0	y_0	$\dot{x}_0$	$\dot{y}_0$	T
0.976995301026691	0	0	0.064369920678510	$T_p \approx 12.475$



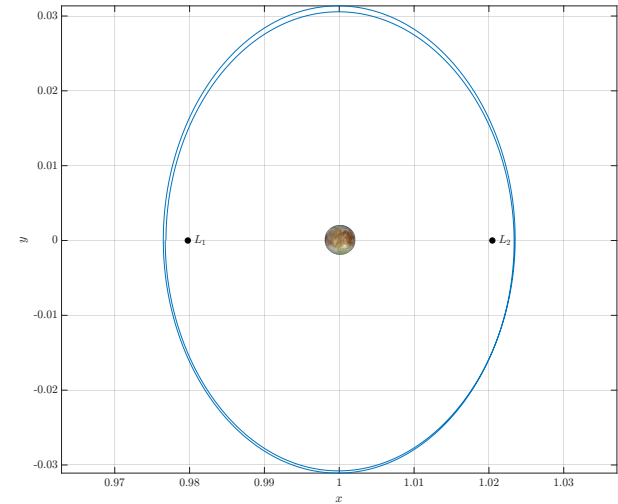
(a) First perpendicular crossing.



(b) Second perpendicular crossing.

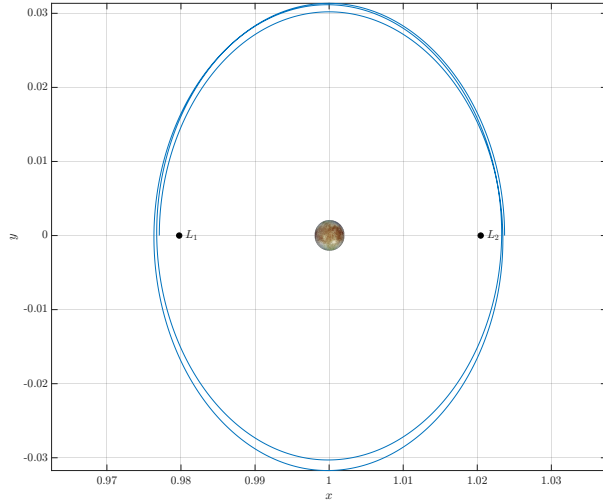


(c) Third perpendicular crossing.

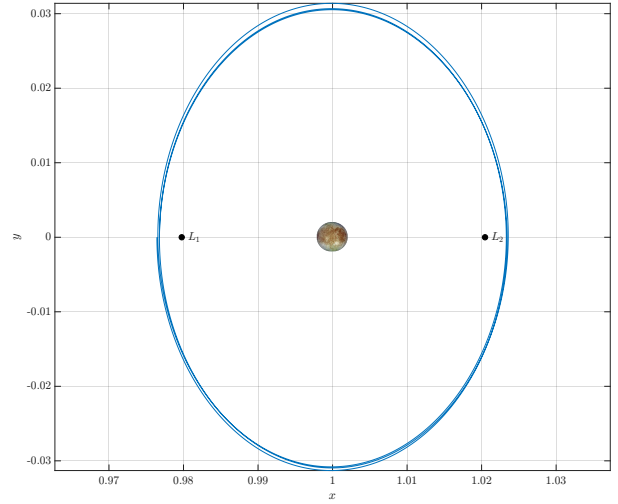


(d) Fourth perpendicular crossing.

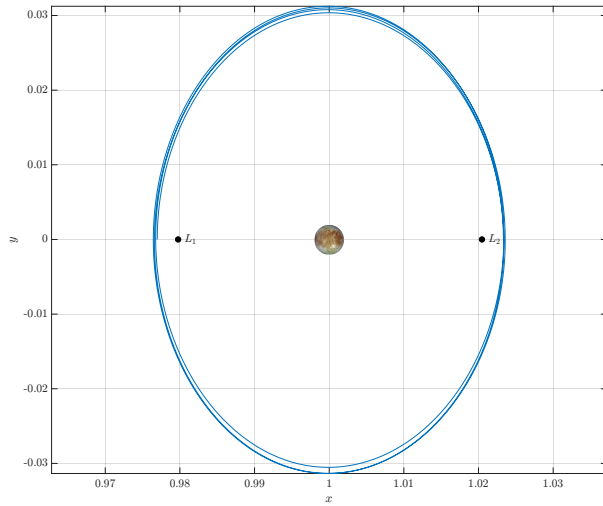
Fig. 10 First four perpendicular crossings needed to generate $\mathcal{P}(T_p, \text{Europa})$.



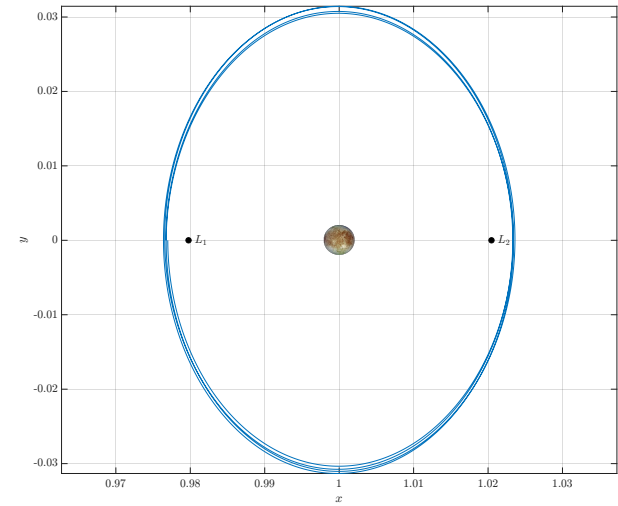
(a) Fifth perpendicular crossing.



(b) Sixth perpendicular crossing.



(c) Seventh perpendicular crossing.



(d) Eighth perpendicular crossing.

Fig. 11 Last four perpendicular crossings needed to generate $\mathcal{P}(T_p, \text{Europa})$

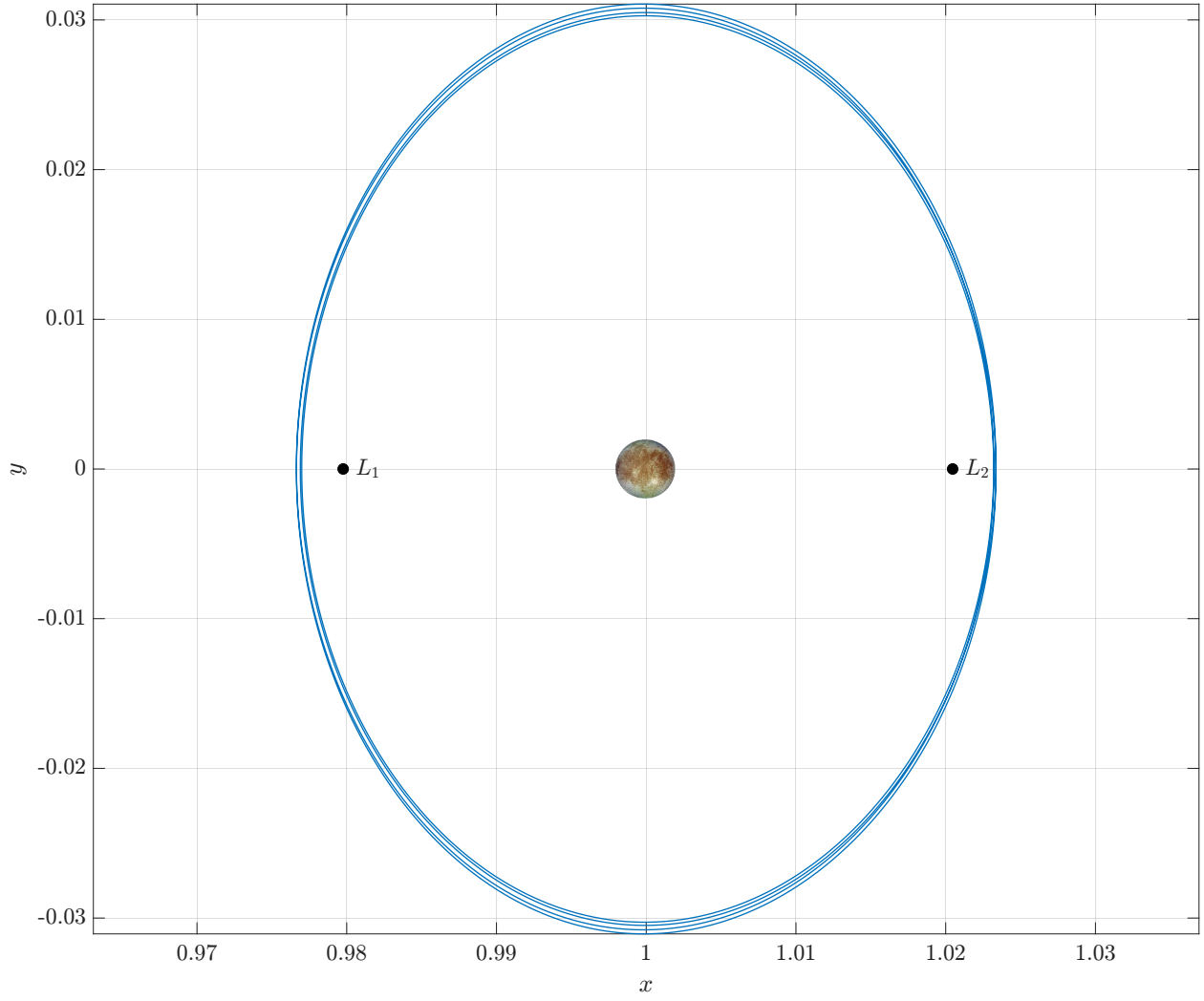


Fig. 12 Periodic Orbit about Europa, $\mathcal{P}(T_p, \text{Europa})$,

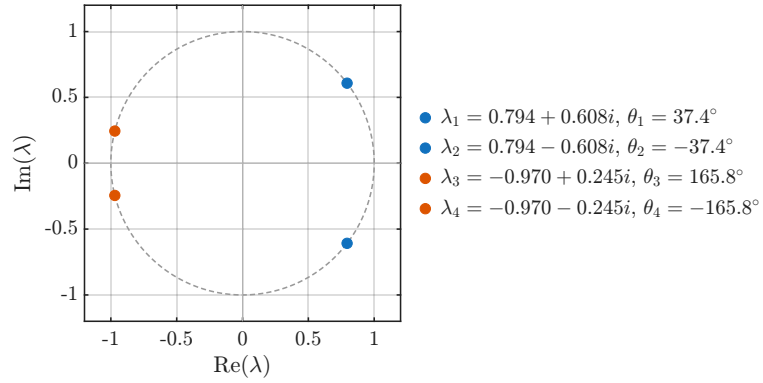


Fig. 13 Eigenvalues of the monodromy matrix $\Phi_M(\mathcal{P}(T_p, \text{Europa}))$. Their magnitude is approximately 1.

G. Interesting Periodic Orbit in the JEG System

We have computed another periodic orbit about L_4 using a standard single-shooting algorithm. It is interesting to note that its period is T_3 , the orbital period of Ganymede.

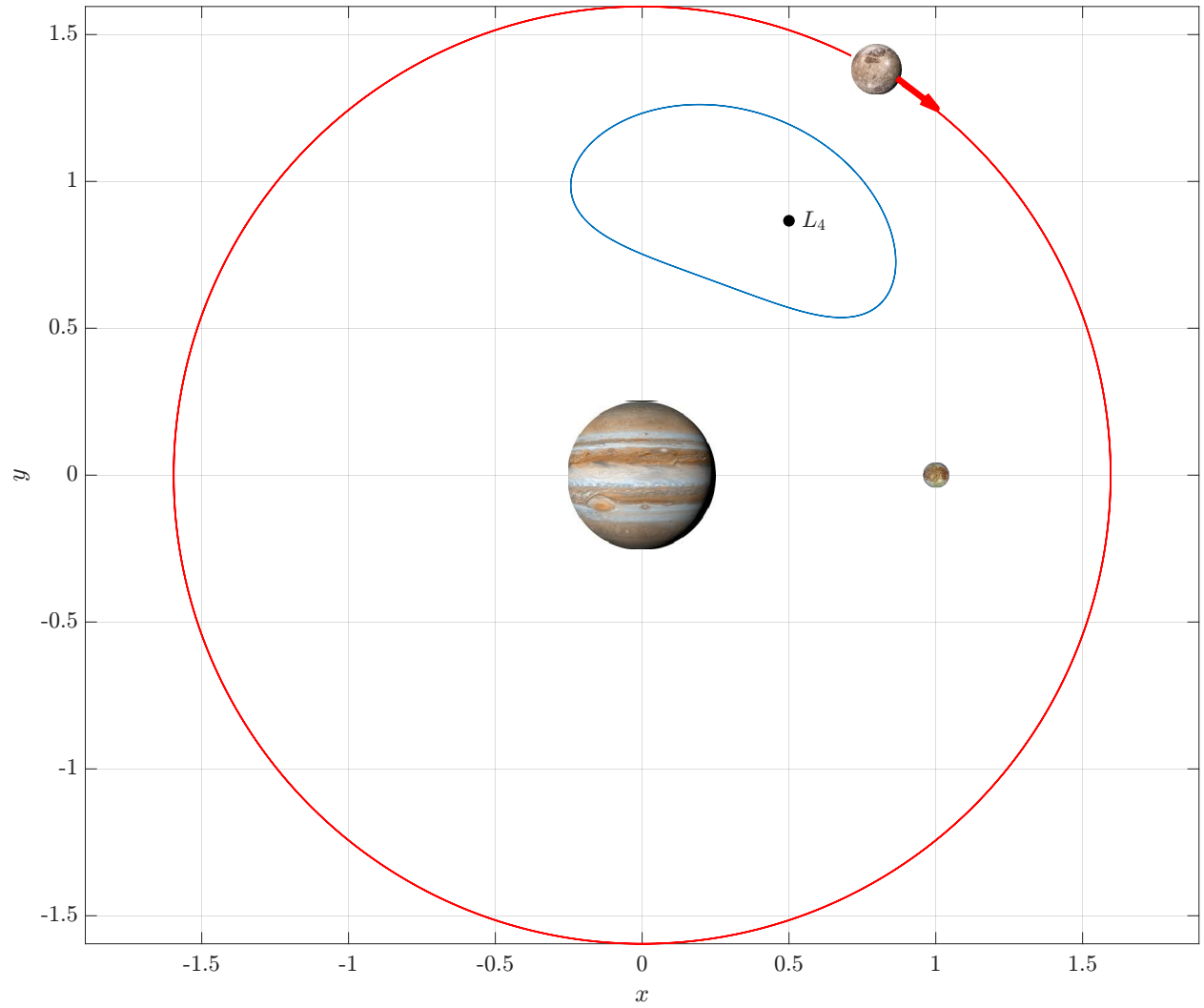


Fig. 14 Periodic Orbit about L_4 . The blue trajectory represents the periodic orbit, the red trajectory represents Ganymede's trajectory.

V. An application to the Saturn - Janus - Epimetheus system

A. System Description

The Saturn–Janus–Epimetheus (SJE) system provides an interesting application of the CCR4BP framework, involving a dominant central mass and two co-orbital moons. Janus and Epimetheus orbit Saturn with nearly identical angular velocities and comparable orbital radii, forming a 1:1 resonance configuration. This setup satisfies the key assumptions of the CCR4BP, namely a dominant central primary and two smaller bodies in nearly circular, concentric orbits. However, unlike the Jupiter–Europa–Ganymede system, the near-equality of the angular rates $\Omega_2 \approx \Omega_3$ leads to a negligible forcing frequency in the synodic frame. As a result, the system is effectively autonomous, offering a useful test case to examine how the CCR4BP behaves in a limiting regime.

Table 5 Parameters for the Saturn–Janus–Epimetheus CCR4BP model.

Parameter	Symbol	Value (Dimensional)	Value (Non-dimensional)
<i>Mass Parameters</i>			
Saturn mass	m_1	5.683×10^{26} kg	$1 - \mu$
Janus mass	m_2	1.98×10^{18} kg	μ
Epimetheus mass	m_3	5.6×10^{17} kg	μ_3
Mass ratio	μ	–	3.48×10^{-9}
Perturbing mass ratio	μ_3	–	9.26×10^{-10}
<i>Janus's Motion</i>			
Saturn–Janus distance	r_{12}	151472 km	1
Janus orbital period	T_2	0.695 days	$2\pi \approx 6.283$
Janus angular rate	Ω_2	$21.58^\circ/\text{hr}$	1
<i>Epimetheus's Motion</i>			
Saturn–Epimetheus distance	r_{13}	151422 km	0.99967
Epimetheus orbital period	T_3	0.695 days	$2\pi \approx 6.283$
Epimetheus angular rate	Ω_3	$21.58^\circ/\text{hr}$	1
Initial phase	θ_0	π	π
<i>Time-Periodic Perturbation</i>			
Forcing period	T_p	–	–

It is important to note that, for these parameters, the CCR4BP reduces to the Circular Restricted Four-Body Problem (CR4BP), in which the three primaries remain fixed on the x -axis and the system becomes autonomous. In this configuration, the dynamical system admits six equilibrium points: four collinear points located on the x -axis and two additional points off the axis [2].

B. Lyapunov Periodic Orbit about L_1

We have computed a periodic orbit about L_1 , denoted $\mathcal{P}(3.594, L_1)$, using a standard single-shooting method. In the CR4BP, L_1 lies to the left of Janus. Two of the four eigenvalues of the corresponding monodromy matrix are equal to unity, reflecting time invariance and energy conservation. The remaining two eigenvalues are real and reciprocal, with magnitudes different from one, indicating the presence of stable and unstable directions. Denoting them by λ_s and λ_u , we have $|\lambda_s| < 1$ and $|\lambda_u| > 1$. Consequently, the motion in the vicinity of $\mathcal{P}(3.594, L_1)$ is unstable.

Table 6 Initial conditions and period of $\mathcal{P}(3.594, L_1)$.

x_0	y_0	$\dot{x}_0$	$\dot{y}_0$	T
-1.000505171383708	0	0	0.000076615010079	3.594

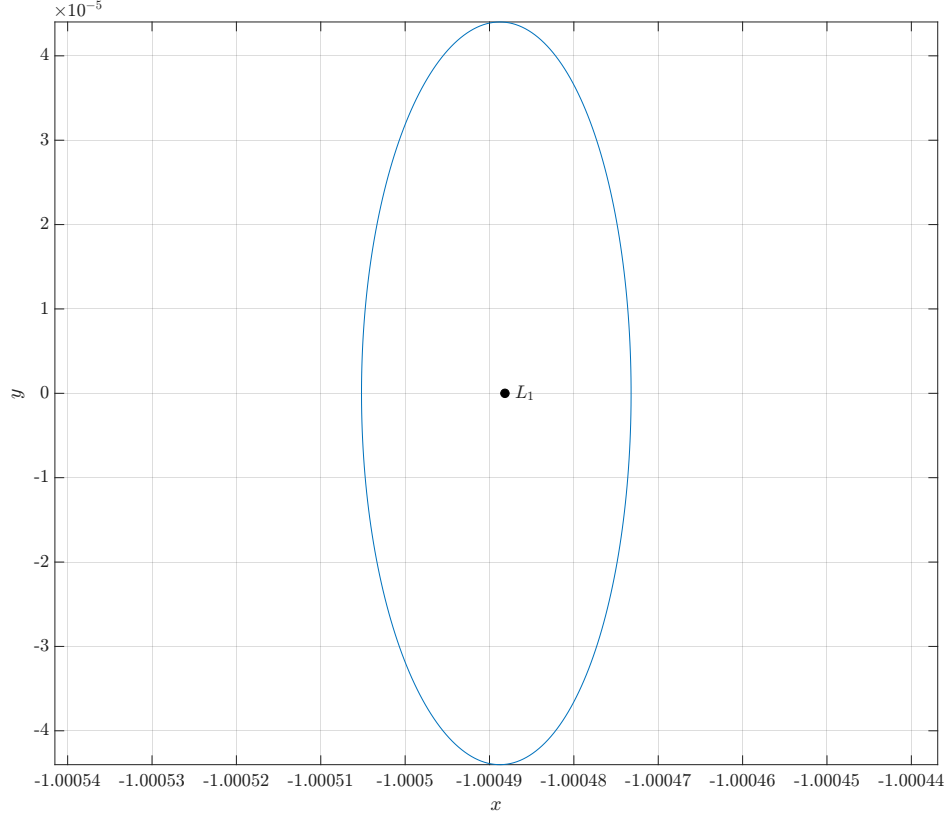


Fig. 15 Periodic Orbit about L_1 of the Saturn - Janus - Epimetheus system.

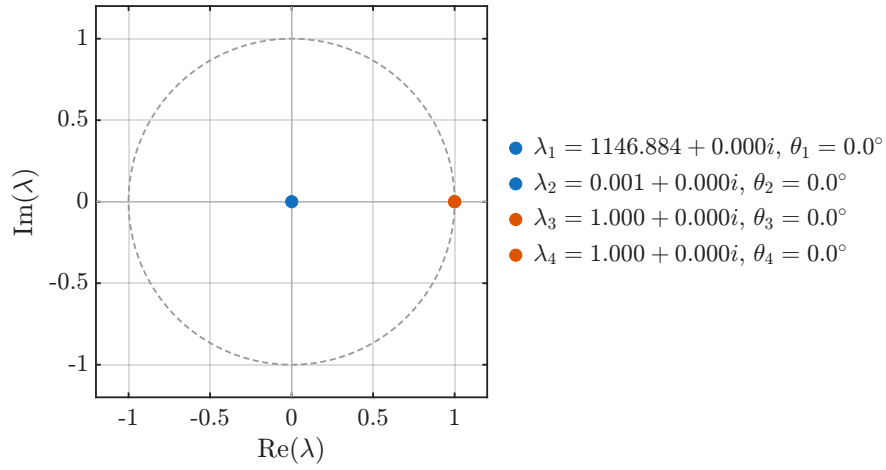


Fig. 16 Eigenvalues of the monodromy matrix $\Phi_M(\mathcal{P}(3.594, L_1))$.

C. Stable and Unstable Manifolds

Consider the STM of $\mathcal{P}(3.594, L_1)$, denoted $\Phi(t, 0)$ for $0 < t < 3.594$. Let $\vec{\omega}_s$ and $\vec{\omega}_u$ be the eigenvectors of the monodromy matrix associated with the stable and unstable eigenvalues λ_s and λ_u , respectively. These directions evolve along the periodic orbit according to

$$\vec{v}_{s,u}(t) = \Phi(t, 0) \vec{\omega}_{s,u}. \quad (31)$$

Let $\varepsilon = 10^{-6}$. Then, the stable and unstable manifolds at each point $\vec{\chi}(t)$ along the orbit can be approximated by the perturbed states

$$\vec{\gamma}_{s,u}^{\pm}(t) = \vec{\chi}(t) \pm \varepsilon \frac{\vec{v}_{s,u}(t)}{|\vec{v}_{s,u}(t)|}. \quad (32)$$

These trajectories are then propagated forward and backward in time to generate the unstable and stable manifolds, respectively, exploiting their invariance under the flow.

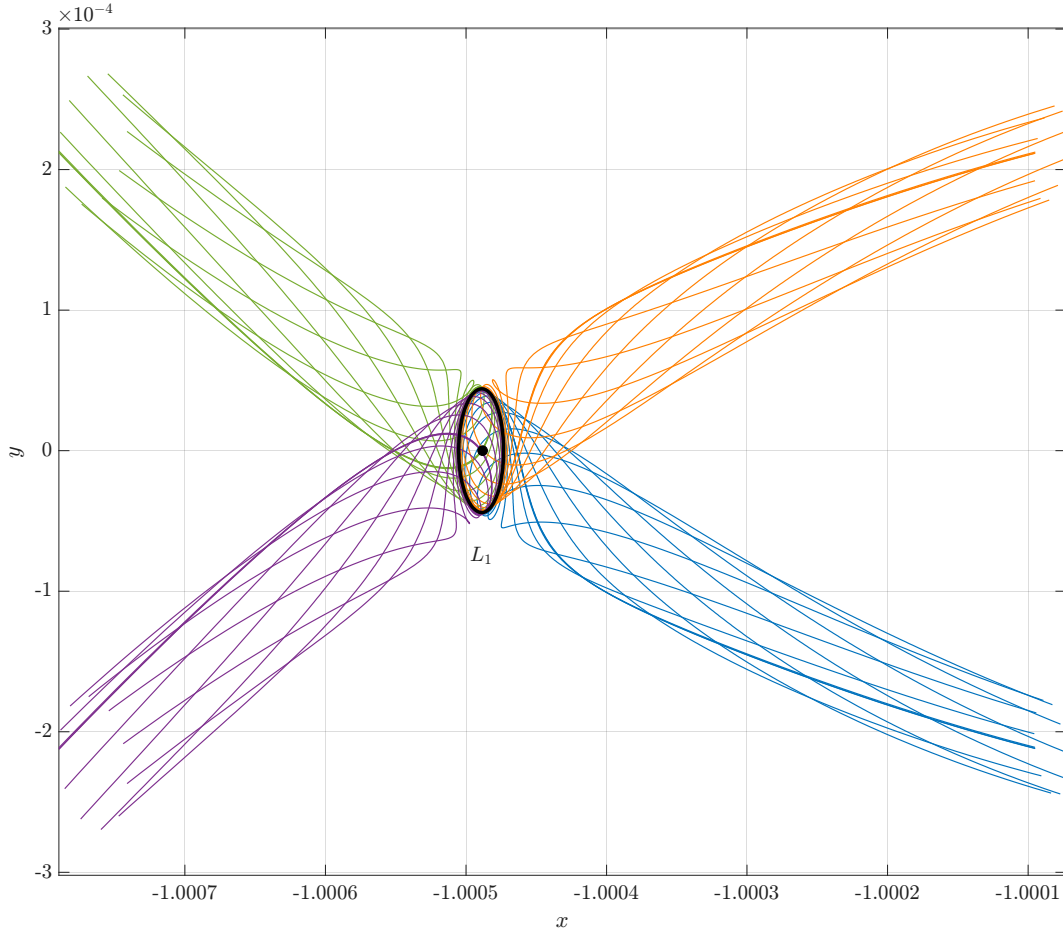


Fig. 17 Stable and unstable manifolds of $\mathcal{P}(3.594, L_1)$ (thick black line) propagated for one orbit period $t = 3.594$. Orange: $\vec{\gamma}_s^+$. Purple: $\vec{\gamma}_s^-$. Blue: $\vec{\gamma}_u^+$. Green: $\vec{\gamma}_u^-$.

VI. Conclusion

In this work, we investigated the dynamical structure of the planar Circular Concentric Restricted Four-Body Problem (CCR4BP), with particular focus on periodic motion in the Jupiter–Europa–Ganymede system. The equations of motion were derived in both Lagrangian and Hamiltonian formulations and integrated together with the State Transition Matrix (STM). Numerical verification through symplecticity checks confirmed that the integrator preserves the Hamiltonian structure with high accuracy. Using both single-shooting and perpendicular crossing techniques, periodic orbits were successfully computed about L_4 , L_5 , and Europa. The analysis of the corresponding monodromy matrices revealed eigenvalues on the unit circle with distinct frequencies, indicating oscillatory stability and the presence of three-dimensional quasi-periodic motion in their vicinity. In contrast, attempts to construct Lyapunov-like periodic orbits about L_1 highlighted the strong instability of this region, which prevented the completion of the required sequence of perpendicular crossings.

On another note, the Saturn–Janus–Epimetheus system was investigated within the CCR4BP framework in the limiting case where the dynamics become effectively autonomous. In this configuration, we computed a periodic orbit about L_1 and propagated its associated stable and unstable manifolds. This demonstrates that the CCR4BP naturally recovers configurations analogous to the classical CR4BP, such as Euler-type collinear arrangements where the three primaries lie on the x -axis.

Future work should focus on improving the computation of unstable periodic orbits, particularly in the vicinity of the collinear Lagrange points, through more robust techniques such as multiple-shooting schemes. Additionally, the computation of the stable and unstable invariant manifolds associated with these orbits would provide deeper insight into the global transport mechanisms of the CCR4BP. Extending the analysis to three-dimensional motion and exploring other resonant configurations within the Jovian or Saturnian system could further reveal complex dynamical structures relevant for trajectory design.

References

- [1] Blazeovski, O. C., D., “Periodic orbits in the concentric circular restricted four-body problem and their invariant manifolds,” *Physica D*, Vol. 241, 2012, pp. 1158–1167. <https://doi.org/10.1016/j.physd.2012.03.008>.
- [2] Michalodimitrakis, M., “The circular restricted four-body problem,” *Astrophysics and Space Science*, Vol. 75, 1981, pp. 289–300.
- [3] Ansari, A. A., “The circular restricted four-body problem with triaxial primaries and variable infinitesimal mass,” *Applications and Applied Mathematics*, Vol. 13, No. 2, 2018, pp. 818–838.
- [4] Idrisi, M. J., Suraj, M. S., Ershkov, S., Abouelmagd, E. I., Beza, D., “A generalized framework for the collinear restricted four-body problem with a central dominant mass,” *Scientific Reports*, Vol. 15, 2025, p. 36440. <https://doi.org/10.1038/s41598-025-20510-y>.
- [5] Kumar, B., Anderson, R. L., de la Llave, R., “Transfers between Jupiter–Ganymede and Jupiter–Europa resonant tori in a concentric circular restricted 4-body model,” *Acta Astronautica*, Vol. 211, 2023, pp. 76–87. <https://doi.org/10.1016/j.actaastro.2023.05.040>.